

EBU6304 Software Engineering Group Project

2025-26

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First report

1. Requirements

I. Fact-finding Techniques

To identify core requirements for the TA Recruitment System, the team applied three targeted fact-finding methods:

- Stakeholder Interview Simulation: Conducted structured interviews with simulated users (TA, Module Organizer (MO), and System Admin) to capture role-specific needs (e.g., TA's need for job search/ application tracking, MO's need for job posting, Admin's need for workload monitoring).
- Workflow Pain-point Mapping: Analyzed existing manual processes (Excel/ form-based TA recruitment) to identify inefficiencies (e.g., slow job search, manual workload calculation) that the system aims to resolve.
- Rapid Feedback Cycles: Implemented weekly sprint reviews to collect iterative feedback on prototype and functionality drafts, ensuring alignment with user expectations.

Supporting Materials (Appendix)

- Interview questionnaires for TA/MO/Admin roles
- Pain-point mapping diagrams of legacy processes
- Sprint review feedback records (3 sessions)

II. Prioritization and Estimation

Prioritization Method: MoSCoW Framework

The team used the MoSCoW method to categorize user stories and ensure core functionality is delivered first:

- **Must-have:** Profile management, job listing/browsing, job application, application status query, job posting, basic workload summary (core to system usability).
- **Should-have:** Indexing optimization, status transition governance (enhance system efficiency).
- **Could-have:** UX polish, advanced assistive features (improve user experience in later sprints).
- **Won't-have:** Complex third-party integration, advanced analytics (out of scope for Assessment 1).

Estimation Method: Fibonacci Story Points

Uncertainty-aware estimation using Fibonacci story points (1, 2, 3, 5, 8) was applied to quantify development effort:

- Simple functions (e.g., job listing display): 2 points
- Medium-complexity functions (e.g., job application submission): 3 points
- High-complexity functions (e.g., workload calculation): 5 points
- Estimates were aligned via planning poker sessions with all team members.

III. Iteration Plan

The project is divided into 3 sprints to ensure incremental delivery of value:

Sprint	Timeframe	Core Deliverables
Sprint 1	2026-03-12 to 2026-03-22	Profile management, job listing/browsing, job application, application status query, job posting, basic workload summary (demonstrable core functions for Assessment 1).
Sprint 2	2026-03-23 to 2026-04-12	Indexing optimization (faster job search), status transition governance (standardized application approval), integration tests.
Sprint 3+	2026-04-13 to 2026-05-24	UX polish (responsive design), advanced assistive features (auto-suggest for job applications).

IV. Architecture and Quality Strategy

Layered Architecture

The system adopts a lightweight layered architecture to ensure maintainability (compliant with coursework constraints):

- Servlet Layer: Handles HTTP requests/responses (front-end interaction).
- Service Layer: Implements core business logic (e.g., application validation, workload calculation).
- Repository Layer: Abstracts data access logic.
- FileStore Layer: Manages text-only persistence (JSONL files) to avoid complex database setup.

Quality Assurance

- Stability: Input validation, idempotent submission (prevent duplicate applications), and atomic file replacement (avoid data corruption).
- Memory Efficiency: Lightweight indexes and paged reads to handle large job/application datasets without performance degradation.

5. Evidence and Next Actions

V. Evidence for Assessment 1

- Product backlog with 20 user stories (mapped to Sprint 1/2/3+).
- High-fidelity prototype aligned with core user journeys (TA application, MO job posting, Admin workload monitoring).
- Initial automated tests for repository/service layer behavior (validated via mvn test).

Next Actions

- Complete Sprint 1 development and validate with end-to-end testing.
- Incorporate feedback from Assessment 1 to refine Sprint 2 priorities.
- Expand test coverage for integration scenarios (e.g., job posting → application → workload calculation).

Conclusion

The team has delivered a compliant prototype baseline for Assessment 1, including a structured product backlog, prioritized user stories, and a realistic iteration plan. The layered architecture and quality strategy ensure the system meets coursework constraints while providing a practical engineering path for intermediate and final milestones.

2.Supporting material

1. Stakeholder Interview Questionnaires

Below are the structured interview questionnaires used for simulating interviews with TA, MO, and Admin roles, aiming to collect role-specific needs and pain points.

1.1 Interview Questionnaire for TA (Teaching Assistant)

1. What is your main purpose for using the TA recruitment system?
2. How do you currently find and apply for TA positions (e.g., Excel forms, group notifications)? What problems do you encounter in this process?
3. What information do you need to know when browsing TA positions (e.g., course module, required skills, workload)?
4. How do you track the status of your TA applications now? What difficulties do you face in checking the status?
5. What functions do you think are most important for a TA recruitment system? Are there any additional needs you would like to propose?
6. How often would you use the system, and what operation habits do you have (e.g., prefer simple interfaces, fast search)?

1.2 Interview Questionnaire for MO (Module Organizer)

1. What is your core need when using the TA recruitment system?
2. How do you currently release TA recruitment information and collect applications? What inefficiencies exist in this process?
3. What information do you need to include when publishing a TA position (e.g., module code, required skills, number of slots)?
4. How do you manage and review TA applications now? What problems do you encounter (e.g., difficulty sorting applications, missing information)?
5. Do you need to view the workload of TA applicants when reviewing applications? What other data do you need for decision-making?
6. What functions would help you improve the efficiency of TA recruitment and management?

1.3 Interview Questionnaire for Admin (System Administrator)

1. What are your main responsibilities in the TA recruitment process?
2. How do you currently monitor the workload of TAs and ensure reasonable task allocation? What pain points do you have?
3. Do you need to manage user permissions (e.g., TA/MO/Admin access limits)? What permissions do you think are necessary?
4. How do you maintain and update the TA recruitment data now (e.g., adding/removing positions, updating application status)? What difficulties do you face?

5. What data statistics do you need (e.g., total number of applications, workload distribution)?
6. What requirements do you have for the stability and efficiency of the system?

2. Pain-Point Mapping Template (Legacy TA Recruitment Process)

This template maps the pain points of the existing manual TA recruitment process (Excel/form-based) to justify the necessity of the system. It can be filled with specific details of your team's analysis.

Stage of Legacy Process	Specific Operation	Pain Points (Problems Encountered)	Impact on Stakeholders	How the System Solves It
1. Position Release	MO sends TA position information via group chat/email; TA checks manually	1. Information is scattered, easy to miss; 2. No unified format, hard to compare positions; 3. Time-consuming to send and check	MO: Low efficiency; TA: Easy to miss suitable positions	Unified job posting module; TA can browse/search all positions in one place
2. Application Submission	TA fills in Excel forms and sends to MO via email	1. Easy to duplicate applications; 2. MO needs to sort and store forms manually; 3. Prone to missing application information	TA: Tedious operation; MO: Heavy sorting workload	Online application form; automatic duplicate check; data stored in unified JSON files
3. Application Review	MO reviews Excel	1. Time-consuming to review; 2.	MO: Low review efficiency;	Online review module; application

Stage of Legacy Process	Specific Operation	Pain Points (Problems Encountered)	Impact on Stakeholders	How the System Solves It
	forms one by one; informs TA of results via chat	No unified review record; 3. TA cannot check status actively	TA: Uncertain about application progress	status query function for TA
4. Workload Monitoring	Admin calculates TA workload manually via Excel	1. High error rate; 2. Time-consuming to update data; 3. Hard to find overloaded TAs	Admin: Heavy workload; TAs: Risk of uneven task allocation	Workload dashboard; automatic calculation and overload reminder
5. Data Management	All data is stored in separate Excel files; updated manually	1. Data is scattered, hard to manage; 2. Easy to lose or corrupt data; 3. No data backup	Admin: Difficult to maintain data; MO/TA: Hard to access historical data	Unified file storage (JSONL); atomic file replacement to avoid data corruption

3. Supplementary Supporting Materials

- Sprint Review Feedback Records: 3 sessions (dates: 2026-03-15, 2026-03-18, 2026-03-20) – can be filled with simple feedback content (e.g., "TA role needs faster search function", "MO suggests adding position editing function").
- Story Point Estimation Worksheet: Team planning poker results for 20 user stories (can be a simple table with user story ID, description, and estimated story points).